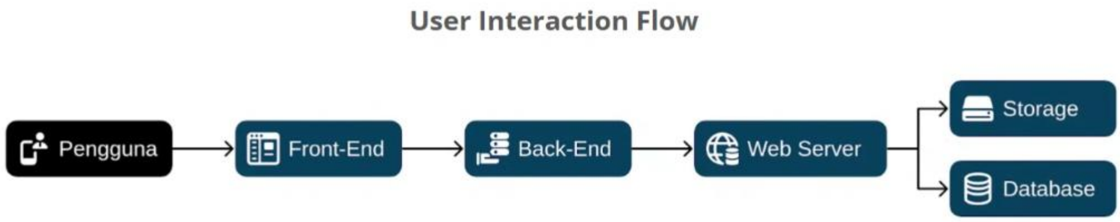
From REST to AI: Bringing Intelligence to Your Back-End

Djoko Imam | Research Lab. Staff

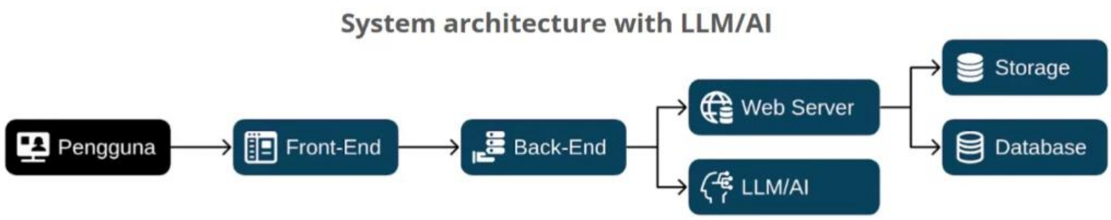
Today's Agenda

1. AI in the Back-End
    - Back-End + AI Application Architecture
2. Building a RESTful API for Local Model Inference
3. Integrating RESTful API with External AI Services
    - Consuming Traditional Models
    - Consuming LLMs
    - Demo!

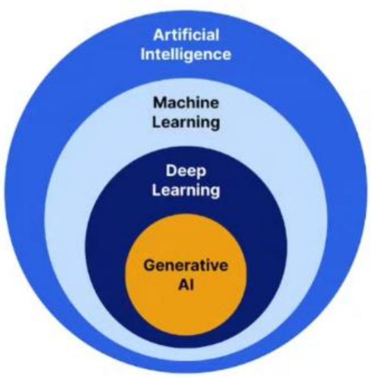
Traditional Application Architecture



Modern Application Architecture



AI, Machine Learning, and Generative AI: An Overview

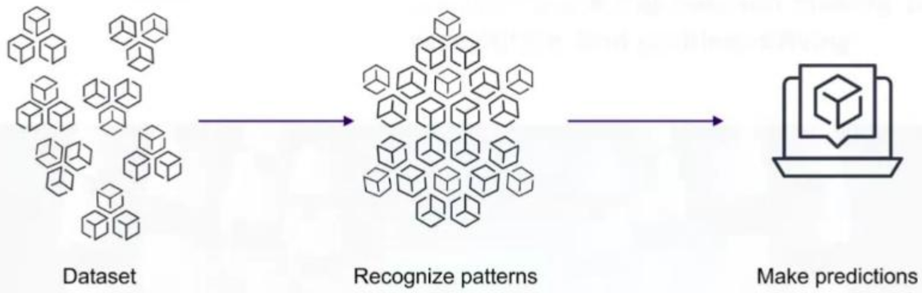


1. *Artificial Intelligence* is the capability of a computer system to perform tasks that require human intelligence.
2. *Machine Learning* are systems capable of learning from data without being explicitly programmed.
3. *Deep Learning* are mathematical models inspired by the structure of the human neural network.
4. *Generative AI* is a branch of AI focused on understanding data to generate new content (text, images, music, video).

An Overview of Artificial Intelligence

Artificial Intelligence (AI) is the core concept behind the entire field of intelligent machines. At its most basic level, AI involves using computers or machines to perform tasks that typically require human intelligence, **such as decision-making, pattern recognition, and problem-solving.**

An Overview of Machine Learning

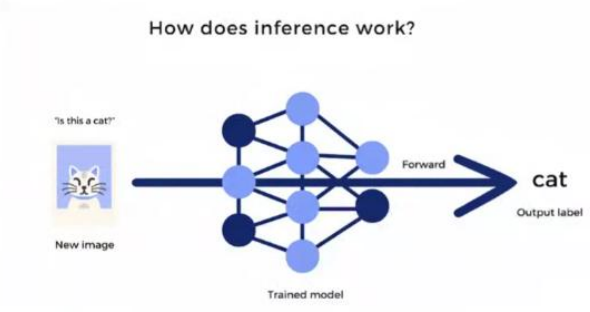


An Overview of Generative AI

Generative AI is a type of AI capable of creating new data based on existing patterns and structures. It can generate various types of content, such as text, images, video, and audio.

Client-side vs. Back-End Inference

Inference is the process of using a pre-trained model to predict outputs based on given inputs.

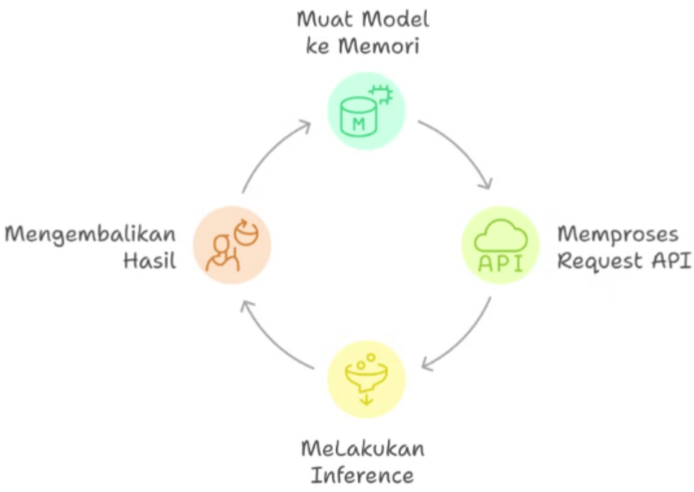


Why Implement AI on the Back-End?

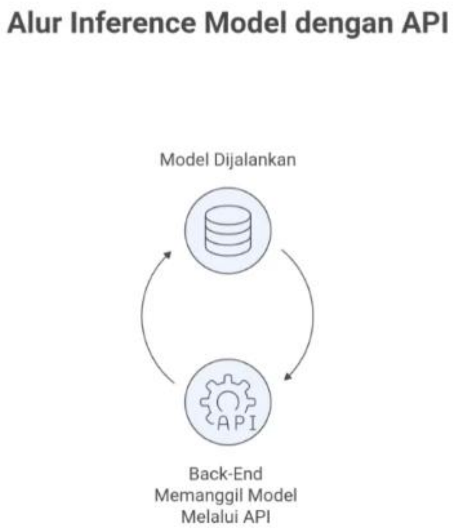
- Data Security
- Stability
- Performance
- Full Control
- Cost Efficiency
- Scalability

Back-End Model Integration

Local Model Inference Pipeline



Model as a Service (MaaS)



Generative AI in the Back-End

Generative AI on the back-end enables the system to generate content (text, images, audio) on-demand via REST API requests.

Building a Simple RESTful API for Local Model Inference

Tech Stack

node JS

Express JS

TensorFlow

What is TensorFlow?

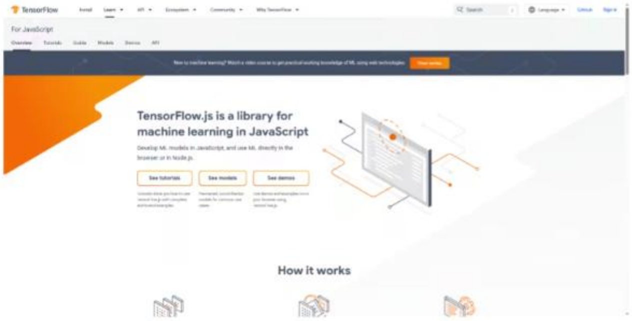
An open-source framework for machine learning and deep learning by Google.

- Used for both model training and inference.
- Supports various model types: classification, NLP, and computer vision.
- Can be executed in Python, C++, and JavaScript.
- Provides export formats ready for REST API integration (e.g., .h5 or .json)

What is TensorFlow.js?

A JavaScript-based version of TensorFlow for running models directly in the browser or Node.js.

- Supports .json and .bin formats.
- Easy to integrate with web applications and REST APIs.




```
const modelPath = `file://${path.resolve(__dirname, '..',
'model', 'model.json')}`;
const model = await tf.loadLayersModel(modelPath);
```


```
const predict = await model.predict(tensor);
const score = await predict.data();
const confidenceScore = Math.max(...score);
const label = tf.argMax(predict, 1).dataSync()[0];

const diseaseLabels = metadata.labels;
const diseaseLabel = diseaseLabels[label];
```

Model Inference: Local vs. Cloud

Aspek	Model Lokal	Model Cloud
Latensi	Cepat (tanpa koneksi internet)	Bergantung koneksi
Sumber Daya	Perlu CPU/GPU lokal	Ditangani cloud
Skalabilitas	Terbatas	Mudah di-scale
Pemeliharaan	Manual	Dikelola provider
Biaya	Murah di awal	Bayar per penggunaan

Activity:
Decision Matrix: Local vs Cloud

Aspek	Model Lokal	Model Cloud
Latensi	Cepat (tanpa koneksi internet)	Bergantung koneksi
Sumber Daya	Perlu CPU/GPU lokal	Ditangani cloud
Skalabilitas	Terbatas	Mudah di-scale
Pemeliharaan	Manual	Dikelola provider
Biaya	Murah di awal	Bayar per penggunaan

Scenario:

A startup needs a **low-cost** solution with **low latency** that does **not rely on stable internet c**onnectivity.

Based on the comparison table, which approach should they choose?

Integrating RESTful APIs with external AI services

Advantages of Using AI Services

- **Unrestricted Access:** Run cutting-edge LLMs without expensive hardware investments.
- **Optimized Spending:** Transform heavy upfront costs into a flexible, pay-as-you-go model.
- **Empowered Developers:** Offload infrastructure management to prioritize product growth and development.

LLMs vs. Conventional AI Models

Karakteristik	Model Tradisional	LLM
Data	Terbatas, domain-spesifik	Dataset besar
Output	Prediksi spesifik	Teks kontekstual
Ukuran	Ringan	Sangat besar (miliar parameter)
Aplikasi	Regresi, klasifikasi	Chatbot, ringkasan, image generation

Cloud Provider LLMs

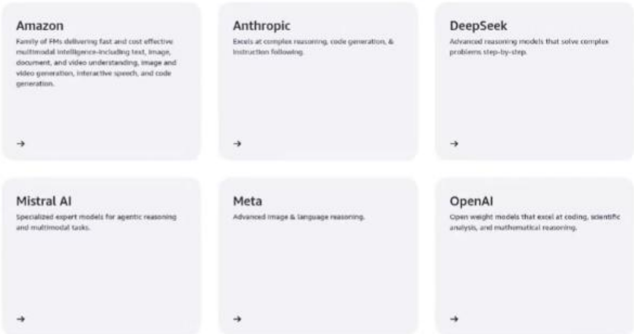
AWS Bedrock

Antropic

Open AI

Vertex AI

Amazon Bedrock: One API, Multiple AI Models



Amazon Bedrock allows you to use foundation models from leading AI providers without needing to manage the underlying infrastructure. You simply choose the best model for your needs and access it via an API.

What is Amazon Bedrock?

AWS Bedrock

Amazon Bedrock is a fully **managed service from AWS** that enables developers to easily build Generative AI applications by leveraging foundation models (FMs).

To use Amazon Bedrock via API, simply install the appropriate SDK for your preferred programming language.

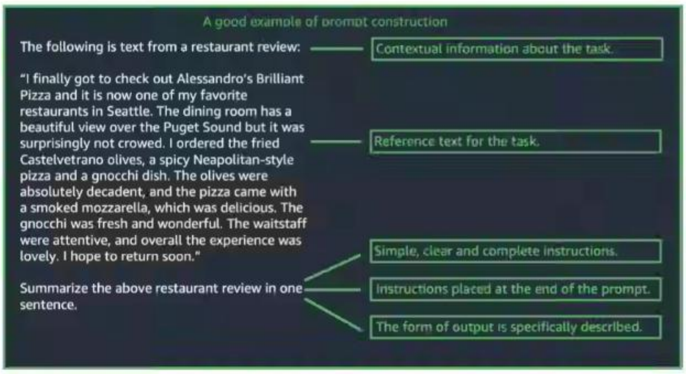
By using Amazon Bedrock, computations are not performed on your local machine, but rather on AWS's global infrastructure.

Tips for Maximizing LLM Utilization

Prompt Engineering

Prompt engineering is the practice of designing and refining the commands (input prompts) you provide to an AI (LLM).

In other words, prompt engineering is the art of communicating with LLMs.



Tips for Maximizing LLM Utilization

System Prompt

A system prompt acts as a foundational instruction that sets the behavior, constraints, and operational context for the AI model before the user interaction begins.

```
[
    {
        "text": "You are an app that creates play lists for a radio station that plays rock and pop music. Only return song names and the artist. "
    }
]
```

Tips for Maximizing LLM Utilization

Token Management & Costs:

Token count affects billing and rate limits (per minute/day).

Use model configurations to monitor and restrict token usage.

```
const request = {
 modelId: this.modelId,
 messages: [message],
 inferenceConfig: {
   maxTokens: config.maxTokens || 500,
   temperature: config.temperature || 0.5,
 },
};
```

Tips for Maximizing LLM Utilization

Inference Parameters

When performing inference, we can adjust the inference parameters to influence the model's response.

- **Temperature** – Controls the degree of randomness in the output generated by the LLM.
- **Top-K** – Controls the creativity of the LLM's output.
- **Top-P** – Controls the diversity and randomness of the LLM's output.

LLM Parameter

Parameter	Low Value Effect (Focused)	High Value Effect (Creative)
Temperature	Outputs are safer, more focused, and predictable.	Outputs are more random, diverse, and creative.
Top K	Limits choices to only the most probable words.	Expands choices to a wider range of words.
Top P	Limits choices to the top words that meet a cumulative probability threshold.	Expands choices to a wider range of words.